Advanced Proof Patterns

Example 1 : Proof by Cases (Three - Way Split)

Prove: For all integers n, $n < 0 \lor n = 0 \lor n > 0$

$$\dfrac{}{n < 0 \lor n = 0 \lor n > 0}\;[\text{trichotomy of integers, axiom}]$$

This uses the trichotomy property as an axiom. To prove something about all integers, we can case-analyze on these three possibilities.

Example 2 : Multi - Level Case Analysis

Prove: $(p \lor q) \land (r \lor s) \Rightarrow (p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p \land r}\,[\land\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor\text{-intro}] \quad \dfrac{\dfrac{}{p \land s}\,[\land\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor \text{ elim on r} \lor \text{s}] \quad \dfrac{\dfrac{\dfrac{}{q \land r}\,[\land\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor\text{-intro}] \quad \dfrac{\dfrac{}{q \land s}\,[\land\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor\text{-intro}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor \text{ elim on r} \lor \text{s}]}{(p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\lor \text{ elim on p} \lor \text{q}]}{((p \lor q) \land (r \lor s)) \Rightarrow (p \land r) \lor (p \land s) \lor (q \land r) \lor (q \land s)}\,[\Rightarrow\text{-intro}^{[1]}]$$

Nested case analysis on two disjunctions, exploring all four combinations.

Example 3 : Proof by Mathematical Induction (Base and Step)

Prove: For all $n \in \mathbb{N}$, sum(1 to n) = n(n+1)/2

Base case ($n = 0$):

$$\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{sum\_to}(0) = 0}\,[\text{definition}]}{0 * (0+1) \ \mathsf{div}\ 2 = 0}\,[\text{arithmetic}]}{\mathit{sum\_to}(0) = 0 * (0+1) \ \mathsf{div}\ 2}\,[\text{equality}]}{\mathit{true} \Rightarrow (\mathit{sum\_to}(0) = 0 * (0+1) \ \mathsf{div}\ 2)}\,[\Rightarrow\text{-intro}^{[1]}]$$

Inductive step (assume for n, prove for n+1):

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner sum\_to(n) = n * (n+1)\ \mathsf{div}\ 2 \urcorner^{[1]}}{sum\_to(n+1) = sum\_to(n) + (n+1)}\ [\text{definition}]}{sum\_to(n+1) = n * (n+1)\ \mathsf{div}\ 2 + (n+1)}\ [\text{substitution}]}{sum\_to(n+1) = (n * (n+1) + 2 * (n+1))\ \mathsf{div}\ 2}\ [\text{algebra}]}{sum\_to(n+1) = (n+1) * (n+2)\ \mathsf{div}\ 2}\ [\text{factoring}]}{(sum\_to(n) = n * (n+1)\ \mathsf{div}\ 2) \Rightarrow (sum\_to(n+1) = (n+1) * (n+2)\ \mathsf{div}\ 2)}\ [\Rightarrow\text{-intro}^{[1]}]$$

By induction, the formula holds for all natural numbers.

Example 4 : Structural Induction on Lists

Prove: For all sequences s, reverse(reverse(s)) = s

Base case (empty sequence):

$$\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner true \urcorner^{[1]}}{reverse(emptyseq) = emptyseq}\ [\text{definition}]}{reverse(reverse(emptyseq)) = reverse(emptyseq)}\ [\text{substitution}]}{reverse(reverse(emptyseq)) = emptyseq}\ [\text{definition}]}{true \Rightarrow (reverse(reverse(emptyseq)) = emptyseq)}\ [\Rightarrow\text{-intro}^{[1]}]$$

Inductive step (assume for s, prove for cons(x, s)):

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner reverse(reverse(s)) = s \urcorner^{[1]}}{reverse(cons(x, s)) = append(reverse(s), x)}\ [\text{definition}]}{reverse(reverse(cons(x, s))) = reverse(append(reverse(s), x))}\ [\text{substitution}]}{reverse(append(reverse(s), x)) = cons(x, reverse(reverse(s)))}\ [\text{definition}]}{cons(x, reverse(reverse(s))) = cons(x, s)}\ [\text{inductive hypothesis}]}{(reverse(reverse(s)) = s) \Rightarrow (reverse(reverse(cons(x, s))) = cons(x, s))}\ [\Rightarrow\text{-intro}^{[1]}]$$

By structural induction, reverse(reverse(s)) = s for all sequences.

Example 5 : Constructive Existence Proof

Prove: There exists an even number greater than 10

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner true \urcorner^{[1]}}{12 = 2 * 6}\ [\text{arithmetic}]}{even(12)}\ [\text{definition of even}, 12 = 2 * 6]}{12 > 10}\ [\text{arithmetic}]}{even(12) \land 12 > 10}\ [\land\text{-intro}]}{\exists\, n : \mathbb{N} \bullet even(n) \land n > 10}\ [\exists\,\text{intro with n = 12}]}{true \Rightarrow (\exists\, n : \mathbb{N} \bullet even(n) \land n > 10)}\ [\Rightarrow\text{-intro}^{[1]}]$$

Constructive proof: we exhibit a specific witness (12).

Example 6 : Non - Constructive Existence Proof

Prove: There exist irrational numbers a and b such that aˆb is rational

$$
\dfrac{
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\dfrac{
\dfrac{}{\mathit{irrational}(\mathit{sqrt}(2)) \land \mathit{irrational}(\mathit{sqrt}(2)) \land \mathit{rational}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)))}\,[\land\text{ -intro}]
}{\exists\, a, b \bullet \mathit{irrational}(a) \land \mathit{irrational}(b) \land \mathit{rational}(\mathit{power}(a, b))}\,[\exists\text{intro with a = b = sqrt(2)}]
\quad
\dfrac{
\dfrac{
\dfrac{
\dfrac{
\dfrac{
\dfrac{}{\mathit{power}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)), \mathit{sqrt}(2)) = \mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2) * \mathit{sqrt}(2))}\,[\text{exponent law}]
}{\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2) * \mathit{sqrt}(2)) = \mathit{power}(\mathit{sqrt}(2), 2)}\,[\text{arithmetic}]
}{\mathit{power}(\mathit{sqrt}(2), 2) = 2}\,[\text{simplification}]
}{\mathit{rational}(2)}\,[\text{known}]
}{\mathit{irrational}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2))) \land \mathit{irrational}(\mathit{sqrt}(2)) \land \mathit{rational}(\mathit{power}(\mathit{power}(\mathit{sqrt}(2), \mathit{sqrt}(2)), \mathit{sqrt}(2)))}\,[\land\text{ -intro}]
}{\exists\, a, b \bullet \mathit{irrational}(a) \land \mathit{irrational}(b) \land \mathit{rational}(\mathit{power}(a, b))}\,[\exists\text{-intro}]
}{\exists\, a, b \bullet \mathit{irrational}(a) \land \mathit{irrational}(b) \land \mathit{rational}(\mathit{power}(a, b))}\,[\lor\text{ -elim}]
}{\mathit{irrational}(\mathit{sqrt}(2)) \Rightarrow (\exists\, a, b \bullet \mathit{irrational}(a) \land \mathit{irrational}(b) \land \mathit{rational}(\mathit{power}(a, b)))}\,[\Rightarrow\text{ -intro}^{[1]}]
$$

Non-constructive: we don't know which case is true, but both lead to the conclusion.

Example 7 : Proof by Strong Induction

Prove: Every natural number $n \geq 2$ has a prime factorization

Base case ($n = 2$):

$$
\dfrac{
\dfrac{
\dfrac{\ulcorner \mathit{true} \urcorner^{[1]}}{\mathit{prime}(2)}\,[\text{definition}]
}{\mathit{prime\_factorization}(2)}\,[\text{trivial, singleton factorization}]
}{\mathit{true} \Rightarrow \mathit{prime\_factorization}(2)}\,[\Rightarrow\text{ -intro}^{[1]}]
$$

Inductive step (assume for all $k < n$, prove for n):

$$
\dfrac{
\dfrac{
\dfrac{}{\mathit{prime\_factorization}(n)}\,[\text{trivial, singleton factorization}]
\quad
\dfrac{
\dfrac{
\dfrac{
\dfrac{
\dfrac{}{\exists\, a, b \bullet 2 \leq a \land a < n \land 2 \leq b \land b < n \land n = a * b}\,[\text{definition of composite}]
}{\mathit{prime\_factorization}(a)}\,[\text{strong IH, a < n}]
}{\mathit{prime\_factorization}(b)}\,[\text{strong IH, b < n}]
}{\mathit{prime\_factorization}(a * b)}\,[\text{multiplication of factorizations}]
}{\mathit{prime\_factorization}(n)}\,[\text{n = a * b}]
}{\mathit{prime\_factorization}(n)}\,[\lor\text{ -elim}]
}{(n \geq 2) \Rightarrow \mathit{prime\_factorization}(n)}\,[\Rightarrow\text{ -intro}^{[1]}]
$$

Strong induction: we assume the property for all smaller values, not just n-1.

Example 8 : Proof Using Lemmas

Lemma 1: If n is even, then nˆ2 is even

$$
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\dfrac{\ulcorner \mathit{even}(n) \urcorner^{[1]}}{\exists\, k \bullet n = 2 * k}\,[\text{definition of even}]
}{\mathit{power}(n, 2) = \mathit{power}(2 * k, 2)}\,[\text{substitution}]
}{\mathit{power}(2 * k, 2) = 4 * \mathit{power}(k, 2)}\,[\text{algebra}]
}{4 * \mathit{power}(k, 2) = 2 * (2 * \mathit{power}(k, 2))}\,[\text{factoring}]
}{\exists\, m \bullet \mathit{power}(n, 2) = 2 * m}\,[\exists\text{intro with m = 2 * power(k, 2)}]
}{\mathit{even}(\mathit{power}(n, 2))}\,[\text{definition of even}]
}{\mathit{even}(n) \Rightarrow \mathit{even}(\mathit{power}(n, 2))}\,[\Rightarrow\text{ -intro}^{[1]}]
$$

Main theorem: If nˆ2 is odd, then n is odd

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{[Lemma1]}
\UnaryInfC{$even(power(n,2))$}
\RightLabel{[contradiction]}
\UnaryInfC{$odd(power(n,2)) \wedge even(power(n,2))$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\RightLabel{[false-elim]}
\UnaryInfC{$odd(n)$}
\AxiomC{}
\RightLabel{[identity]}
\UnaryInfC{$odd(n)$}
\RightLabel{[$\vee$ -elim]}
\BinaryInfC{$odd(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$odd(power(n,2)) \Rightarrow odd(n)$}
\end{prooftree}
$$

Proof by contrapositive using lemma.

Example 9 : Proof by Minimal Counterexample

Prove: All natural numbers $n \geq 1$ satisfy $p(n)$

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{[base case proved separately]}
\UnaryInfC{$p(1)$}
\RightLabel{[contradiction]}
\UnaryInfC{$\neg\,((p(m))) \wedge p(1)$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\AxiomC{}
\RightLabel{[minimality of m]}
\UnaryInfC{$\forall\, k \bullet k \geq 1 \wedge k < m \Rightarrow p(k)$}
\RightLabel{[since m $-$ 1 >= 1 $\wedge$ m $-$ 1 < m]}
\UnaryInfC{$p(m-1)$}
\RightLabel{[by inductive step from p(m $-$ 1)]}
\UnaryInfC{$p(m)$}
\RightLabel{[contradiction]}
\UnaryInfC{$\neg\,((p(m))) \wedge p(m)$}
\RightLabel{[contradiction]}
\UnaryInfC{false}
\RightLabel{[$\vee$ -elim]}
\BinaryInfC{false}
\RightLabel{[$\neg$ -intro$^{[2]}$]}
\UnaryInfC{$\forall\, n \bullet n \geq 1 \Rightarrow p(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$\mathit{true} \Rightarrow (\forall\, n \bullet n \geq 1 \Rightarrow p(n))$}
\end{prooftree}
$$

Minimal counterexample combines well-ordering with contradiction.

Example 10 : Proof by Invariant

Prove: A loop maintains invariant I

Initialization:

$$
\begin{prooftree}
\AxiomC{$\ulcorner \mathit{initial\_state} \urcorner^{[1]}$}
\RightLabel{[verification]}
\UnaryInfC{$\mathit{invariant}(\mathit{initial\_state})$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$\mathit{initial\_state} \Rightarrow \mathit{invariant}(\mathit{initial\_state})$}
\end{prooftree}
$$

Preservation:

$$
\begin{prooftree}
\AxiomC{$\ulcorner \mathit{invariant}(\mathit{before\_state}) \wedge \mathit{executes\_loop\_body} \urcorner^{[1]}$}
\RightLabel{[$\wedge$ -elim-1]}
\UnaryInfC{$\mathit{invariant}(\mathit{before\_state})$}
\RightLabel{[$\wedge$ -elim-2]}
\UnaryInfC{$\mathit{executes\_loop\_body}$}
\RightLabel{[verification]}
\UnaryInfC{$\mathit{invariant}(\mathit{after\_state})$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$(\mathit{invariant}(\mathit{before\_state}) \wedge \mathit{executes\_loop\_body}) \Rightarrow \mathit{invariant}(\mathit{after\_state})$}
\end{prooftree}
$$

Termination:

$$\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner loop\_terminates \urcorner^{[1]}}{invariant(termination\_state)}\ \text{[by preservation]}}{invariant(termination\_state) \wedge termination\_condition}\ [\wedge\text{-intro}]}{desired\_property}\ \text{[logic]}}{loop\_terminates \Rightarrow desired\_property}\ [\Rightarrow\text{-intro}^{[1]}]$$

Example 11 : Proof by Diagonalization

Prove: The set of real numbers is uncountable

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner countable(reals) \urcorner^{[2]}}{\exists f \bullet enumeration(f, reals)}\ \text{[definition of countable]}}{diagonal\_construction(r)}\ \text{[diagonal method]}}{\forall n \bullet r \neq apply(f, n)}\ \text{[by construction, differs at nth digit]}}{not\_in\_range(r, f)}\ \text{[previous line]}}{not\_in\_range(r, f) \wedge enumeration(f, reals)}\ \text{[contradiction]}}{false}\ \text{[contradiction]}}{uncountable(reals)}\ [\neg\text{-intro}^{[2]}]}{true \Rightarrow uncountable(reals)}\ [\Rightarrow\text{-intro}^{[1]}]$$

Cantor's diagonal argument (outline).

Example 12 : Constructive Proof Pattern

To constructively prove: $\exists x \bullet p(x)$

Strategy: 1. Explicitly construct a witness w 2. Verify $p(w)$ holds 3. Conclude $\exists x \bullet p(x)$ with $x = w$

Example: Prove $\exists n : \mathbb{N} \bullet n > 100 \wedge even(n)$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\ulcorner true \urcorner^{[1]}}{witness\_construction(102)}\ \text{[construction]}}{102 > 100}\ \text{[arithmetic]}}{102 = 2 * 51}\ \text{[arithmetic]}}{even(102)}\ \text{[definition]}}{102 > 100 \wedge even(102)}\ [\wedge\text{-intro}]}{\exists n \bullet n > 100 \wedge even(n)}\ [\exists\text{intro with n} = 102]}{true \Rightarrow (\exists n \bullet n > 100 \wedge even(n))}\ [\Rightarrow\text{-intro}^{[1]}]$$

Example 13 : Proof Composition

Combine multiple proof techniques:

Theorem: Property P holds for all cases

Overall strategy: Case analysis + Induction + Contradiction

$$
\begin{prooftree}
\AxiomC{}
\RightLabel{[direct proof]}
\UnaryInfC{$p(base)$}
\AxiomC{$\ulcorner \neg\ ((p(n)))\urcorner^{[2]}$}
\RightLabel{[proof steps]}
\UnaryInfC{$contradiction\_derived$}
\RightLabel{[contradiction]}
\UnaryInfC{$false$}
\RightLabel{[$\neg$ -intro$^{[2]}$]}
\UnaryInfC{$p(n)$}
\AxiomC{$\ulcorner p(n-1)\urcorner^{[3]}$}
\RightLabel{[inductive step]}
\UnaryInfC{$p(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[3]}$]}
\UnaryInfC{$p(n)$}
\RightLabel{[$\vee$ elim over subcases]}
\BinaryInfC{$p(n)$}
\RightLabel{[$\vee$ elim over cases]}
\BinaryInfC{$\forall\, n \bullet p(n)$}
\RightLabel{[$\Rightarrow$ -intro$^{[1]}$]}
\UnaryInfC{$true \Rightarrow (\forall\, n \bullet p(n))$}
\end{prooftree}
$$

Example 14 : Best Practices for Complex Proofs

Guidelines for writing advanced proofs:

1. State strategy at the beginning 2. Label cases clearly 3. Discharge assumptions promptly 4. Reference lemmas explicitly 5. Show key algebraic steps 6. Justify non-obvious steps 7. Use proof by cases when structure suggests it 8. Use induction for recursive definitions 9. Use contradiction for negative conclusions 10. Verify base cases thoroughly

Example 15 : Proof Documentation

Document complex proofs:

- **Goal**: State what you're proving - **Strategy**: Explain the proof approach - **Lemmas needed**: List dependencies - **Key insights**: Highlight non-obvious steps - **Pitfalls**: Note where proof could go wrong - **Generalization**: Explain how proof extends

Well-documented proofs are maintainable and reusable.